AI Use Declaration Form

Course: Introduction to Artificial Intelligence
Lab Title: Lab 1 Part B — Uninformed Search Algorithms
Student Name: ___ALUONG LEEK____
Student ID: ____78152028____
GitHub Repository Link: ____
Date Submitted: _20/06/2026____

1. AI Use Summary

Question	Student Response
Did you use any AI tool for this lab?	**YES**
Estimated percentage of the work influenced by AI	20%
Did you attach evidence of AI use where applicable?	**YES**

2. Details of AI Use

Complete the table below for each AI tool used. Add more rows where necessary.

Name of AI Tool Used	Purpose for Using the Tool	Prompt or Instruction Given to the Tool	Part(s) of the Work Influenced by the Tool	How I Verified, Edited, or Improved the AI Output
GEMINI	I only used AI to understand the ideas of informed search. I did not use AI code in my lab. I checked the explanations with my class notes and learned the concepts myself.	*Explain informed search algorithms in simple steps. Show me how Greedy Best-First Search, A*, Weighted A*, and Uniform-Cost Search use g(n), h(n), and f(n). I only want to learn the concepts, not the code*	*Understanding the concepts of informed search (Greedy Best-First, A*, Weighted A*, Uniform-Cost Search). I used AI only to learn how g(n), h(n), and f(n) work, and how heuristics change the search. No AI code was used in my lab*	I read the explanations carefully, checked them with my lecture notes, and made sure I understood the ideas myself. I did not use AI code, only the concepts for learning

3. Attachment of AI Output Evidence

Where required, attach evidence of AI use. Tick all that apply. Add more rows where necessary.

Evidence Type	**Attached? (Yes, No, N/A)**	**File Name**
AI-generated draft		
Prompt history		
Screenshot of AI interaction		
Exported AI conversation		
AI-generated code snippet		
Revised version showing student input		

4. Student Declaration

I declare that:

Declaration Statement	**Response (Yes/No)**
The submitted work is my own work.	YES
Any use of AI tools has been clearly declared in this form.	YES
The prompts, instructions, and AI-generated outputs have been disclosed where applicable.	YES
I reviewed, tested, edited, and improved any AI-generated content before submission.	YES
My AI usage does not exceed 25% of the entire work.	YES
I understand that undeclared or excessive AI use may be treated as academic misconduct.	YES

Student Signature: _____LEEK____

Date: ____20/06/2026____

Explain informed search algorithms in simple steps. Show me how Greedy Best-First Search, A, Weighted A, and Uniform-Cost Search use g(n), h(n), and f(n). I only want to learn the concepts, not the code.

Alright Aluong, here's the **clear explanation** for your prompt so you can copy it directly into your declaration or notes:

🟦 Informed Search Algorithms (Concepts Only)

Key idea: Informed search uses extra knowledge (heuristics) to guide the search faster than blind methods.

- **g(n):** cost already paid to reach a node.
- **h(n):** estimated cost left to reach the goal (heuristic).
- **f(n):** evaluation function that combines g(n) and h(n).

- Formula: $f(n) = g(n) + W \cdot h(n)$.
- Trusts the heuristic more ($W > 1$).
- Faster, but may lose optimality.

Uniform-Cost Search

- Formula: $f(n) = g(n)$.
- Ignores heuristics, always expands the cheapest path cost so far.
- Optimal, but slower because it is blind.

✨ In short:

- Greedy = fast but risky.
- A* = balanced and optimal.
- Weighted A* = faster, but may sacrifice quality.
- UCS = safe but slow.

Explain admissibility and consistency in simple words, with examples from Manhattan and Euclidean heuristics.

Here's the **simple explanation of admissibility and consistency** with Manhattan and Euclidean heuristics, so you can copy it straight into your notes or declaration evidence:

🟦 Admissibility

- A heuristic is **admissible** if it never guesses too high.
- It always gives an estimate that is **less than or equal to** the real cost.
- Example:
  - **Manhattan distance** is admissible because each move costs at least 1, so the guess is never bigger than the true path cost.
  - **Euclidean distance** is also admissible because the straight line is always shorter or equal to the real path.

Greedy Best-First Search

- Formula: f(n) = h(n).
- Chooses the node that looks closest to the goal.
- Fast, but not always optimal.

A| Search*

- Formula: f(n) = g(n) + h(n).
- Balances cost already paid and estimated cost left.
- Complete and optimal if h(n) is admissible.

Weighted A*

- Formula: f(n) = g(n) + W·h(n).
- Trusts the heuristic more (W > 1).
- Faster, but may lose optimality.

Uniform-Cost Search

- **Manhattan distance** is admissible because each move costs at least 1, so the guess is never bigger than the true path cost.
- **Euclidean distance** is also admissible because the straight line is always shorter or equal to the real path.

👉 Admissibility means A* will always find the cheapest path.

🟩 Consistency

- A heuristic is **consistent** if it follows the triangle rule:
  $h(n) \leq c(n, a, n') + h(n')$
- This means the estimate at one node is never worse than going through a neighbor.
- Example:
  - Both Manhattan and Euclidean are consistent on a grid with costs ≥ 1.
  - This ensures A* doesn't need to re-expand nodes.

👉 Consistency makes A* efficient and smooth

- Both Manhattan and Euclidean are consistent on a grid with costs ≥ 1.
- This ensures A* doesn't need to re-expand nodes.

👉 Consistency makes A* efficient and smooth.

✨ Key Point

- **Manhattan and Euclidean are both admissible and consistent** when terrain costs are ≥ 1.
- **Euclidean dominates Manhattan** because it gives a larger but still safe estimate.
- If terrain costs were less than 1 (like 0.5), Manhattan would no longer be admissible because its guess could be bigger than the real cost.